


Exam 1 on Friday

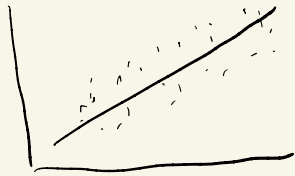
- around 12-14 questions
- a couple multiple choice, and the rest is short answer
- you do not need to write any R code
- bring a calculator!

Regression assumptions

- Shape : the true relationship is close enough to the chosen model

e.g. If model is $y = \beta_0 + \beta_1 x + \varepsilon$
 $\Rightarrow$ assuming linear relationship between x & y

- $\varepsilon \sim N(0, \sigma^2)$ $\swarrow$ constant variance ; variance of noise term ε is constant (variability of ε doesn't depend on x)
 $\nwarrow$ normality ;

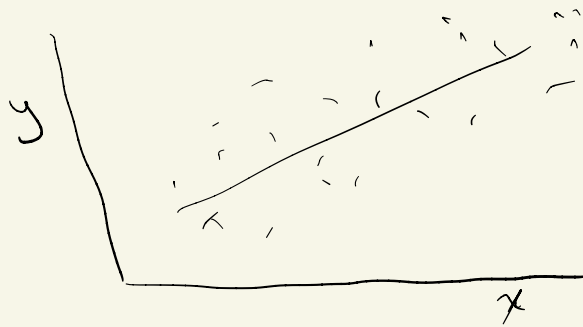


$$y = \beta_0 + \beta_1 x + \varepsilon$$

$\uparrow$ noise (aka random error)

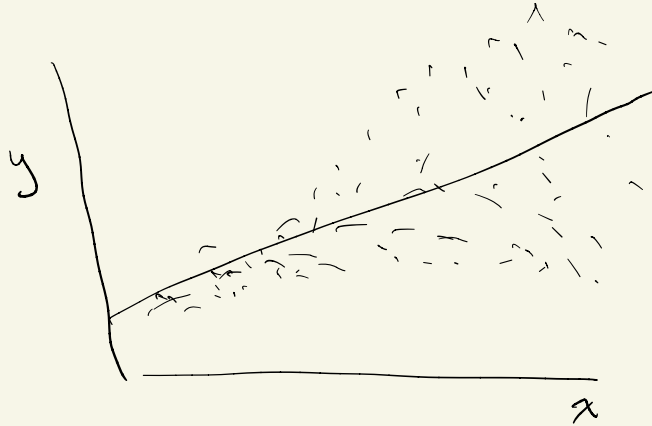
normality assumption : $\varepsilon \sim \text{Normal}$

- independence : observations are all independent
choosing something about one point doesn't influence another point
- randomness : data come from random process
(e.g., random sample, randomized experiment)



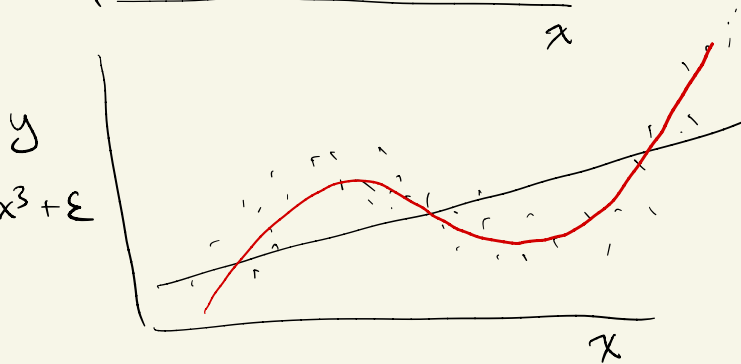
Shape ✓

constant variance ✓



Shape ✓

constant variance ✗

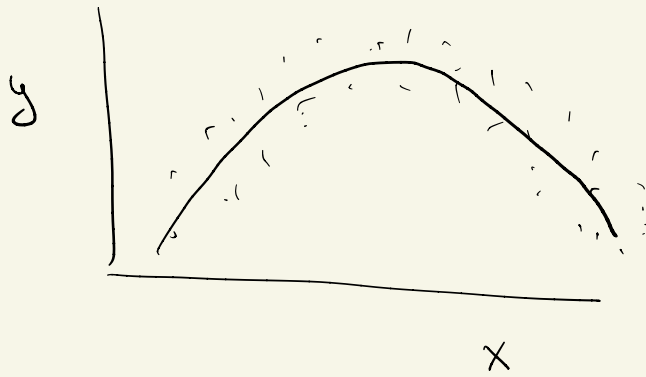


constant variance ✓

shape ✗

→ maybe use
3rd order
polynomial?

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \varepsilon$$

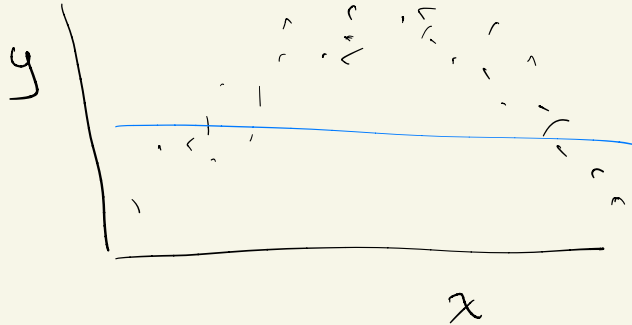


model: $y = \beta_0 + \beta_1 x + \beta_2 x^2 + \varepsilon$

Shape assumption: true

relationship is quadratic

Here Shape assumption ✓



model: $y = \beta_0 + \beta_1 x + \varepsilon$

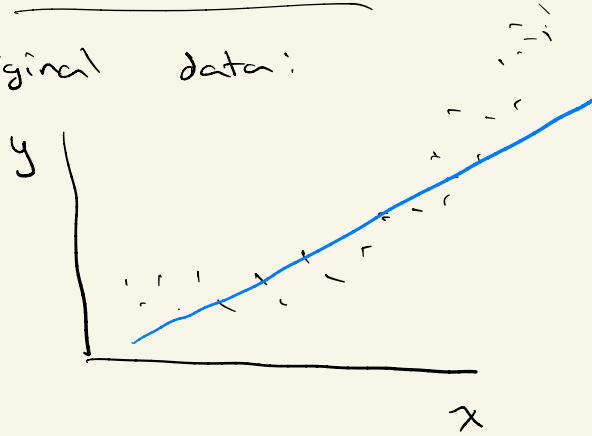
Shape assumption: true

relationship is linear

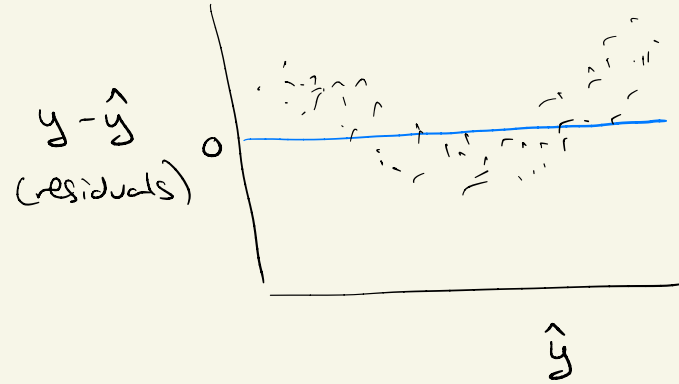
Here shape assumption
is not satisfied

residual plot

original data:



residual plot:



pattern in the residual plot
(e.g., U-shaped pattern)
suggests shape assumption
is not satisfied

QQ plot

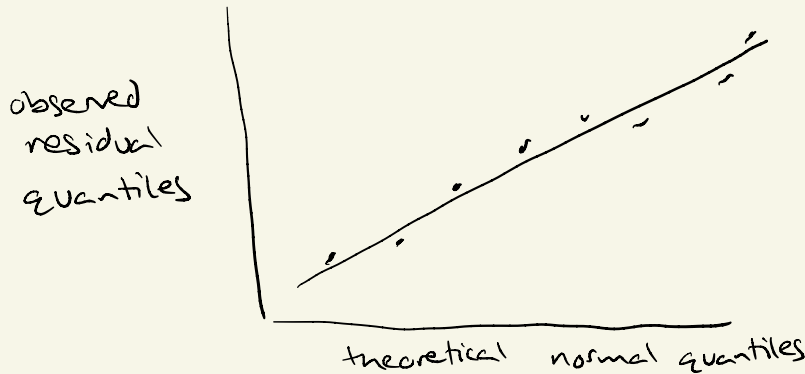
(quantile - quantile plot)

Normality

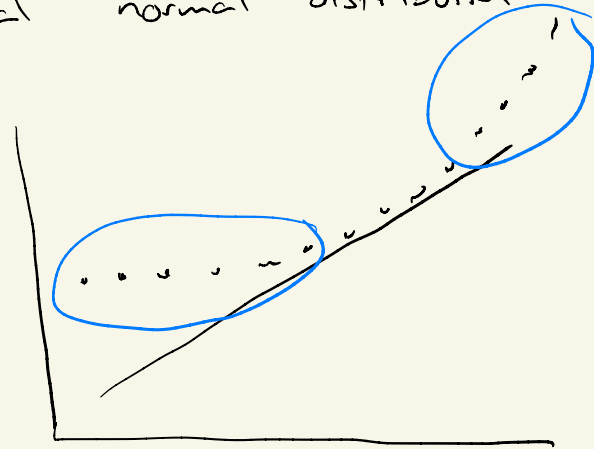
assumption : $\varepsilon \sim \text{Normal}$

$$\Rightarrow \underset{\substack{\uparrow \\ \text{observed}}}{y_i} - \underset{\substack{\uparrow \\ \text{predicted}}}{\hat{y}_i} \approx \text{Normal}$$

So: compare quantiles of observed residuals to quantiles of theoretical normal distribution



Normal residuals
(normality ✓)



Non-normal residuals
(normality ✗)

Notation:

observed data: $(x_1, y_1), \dots, (x_n, y_n)$ (n observations)

x_i = value of explanatory variable for i^{th} observation

y_i = value of response variable for i^{th} observation

fitted model:

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

↑ ↑ ↑
predicted value estimated coefficients
of response

hat (^) : always a predicted / estimated value

mean of y : $\bar{y}$ denotes average value of response variable in data

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

bar (-) : an average

Population model:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

Fitted model:

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$$

- hats

- no ε

On exam:

replace X & Y w/ actual variable names

replace $\hat{\beta}_0, \hat{\beta}_1$ w/ actual estimates
from R output

Ex: penguins data, relationship between bill length & bill depth

$$\text{bill depth} = \beta_0 + \beta_1 (\text{bill length}) + \varepsilon$$

$$\hat{\text{bill depth}} = 1.78 + 0.22 (\text{bill length})$$

Last day of material for Exam 1: Feb 16
(up to inference & diagnostics)

Hypothesis testing: kinds of questions I might ask could include

• write down H_0 and H_A in terms of model parameter

e.g., $H_0: \beta_1 = 0$ $H_A: \beta_1 < 0$

$$H_0: \beta_1 = 0 \quad H_A: \beta_1 \neq 0$$

$$H_0: \beta_1 = 3 \quad H_A: \beta_1 \neq 3$$

• calculate a test statistic
$$t = \frac{\hat{\beta}_1 - (\text{value under } H_0)}{SE\hat{\beta}_1}$$

e.g. $H_0: \beta_1 = 0 \rightarrow t = \frac{\hat{\beta}_1 - 0}{SE\hat{\beta}_1}$

• what distribution & degrees of freedom?

Under H_0 , $t \sim t_{n-2} \leftarrow \begin{array}{l} n \text{ observations} \\ 2 \text{ parameters } (\beta_0, \beta_1) \end{array}$

more generally: $t \sim t_{n-p}$

$p = \#$ of β s in model

• Shade area corresponding to p-value for test

t_{n-2}

eg. $t = 1.8$

$H_0: \beta_1 = 0$

$H_A: \beta_1 \neq 0$

